Viva Preparation

Experiments 4, 5, and 6

Data Science Lab | Hypothesis testing, HDFS, MapReduce Word Count | Ojas Srivastava, U24AI028

How to answer

State the aim, what you did on the Titanic or Hadoop setup, your numbers or command, then the conclusion. For statistics: H0, H1, test used, p-value, alpha = 0.05, reject or fail to reject.

Experiment 4: Statistical hypothesis testing

Aim

Test claims about the Titanic data using Python and decide with p-values (alpha = 0.05).

Dataset

Titanic, 891 passengers. Main variables: Age, Fare, Survived, Sex. Rows with missing Age were dropped for the age test.

Key ideas

- H0: no difference / no association. H1: there is a difference or association.
- p-value: how surprising your result is if H0 is true. If p < 0.05, reject H0.
- Independent t-test: compare means of two groups (survived vs not survived). Use when spread in the population is unknown; use sample s (Welch t-test, equal_var=False in SciPy).
- Chi-square: test link between two categories (Sex vs Survived).
- t-test vs z-test: unknown variance and small/medium sample -> t. Known population variance or large-n normal approximation -> z. This lab uses t.

Main result (Age)

- H0: mean age same for survived and not survived. H1: means differ.
- Test: scipy.stats.ttest_ind(..., equal_var=False)
- p-value = 0.041 (< 0.05) -> Reject H0.
- Conclusion: mean age differs significantly between survived and not-survived passengers.

Other results from the notebook

- Fare t-test: p about 2.7e-11 -> reject H0; mean fare differs by survival.
- Sex vs Survived chi-square: chi2 about 260.7, p about 1.2e-58 -> reject H0; sex and survival are associated.

Exercises

Q: 1. t-test on Fare instead of Age.

A: Same two groups. Fare p-value is very small; reject H0. Survivors paid higher fares on average.

Q: 2. Change alpha from 0.05 to 0.01 (Age test).

A: p = 0.041. Still reject at 0.05; fail to reject at 0.01 (0.041 is not < 0.01).

Q: 3. Interpret the p-value.

A: Probability of a difference at least this large if H0 were true. Small p -> evidence against H0.

Q: 4. Chi-square on Sex vs Survived.

A: Build a contingency table; stats.chi2_contingency. Strong association; reject independence.

Q: 5. Hypotheses for Fare.

A: H0: equal mean fare. H1: unequal mean fare. Same as exercise 1.

Viva questions

Q: What is H0?

A: No significant difference or no association.

Q: When reject H0?

A: When p-value < alpha (here 0.05).

Q: What is alpha?

A: Significance level; risk of false alarm. Usually 0.05.

Q: Independent t-test?

A: Compares means of two separate groups.

Q: Chi-square test?

A: Checks if two categorical variables are independent.

Q: What library?

A: SciPy stats: ttest_ind, chi2_contingency.

Experiment 5: Hadoop and HDFS

Aim

Install/configure Hadoop on one machine and run basic HDFS file commands.

Components

- HDFS stores data in blocks on DataNodes. NameNode holds metadata only.
- YARN schedules jobs. MapReduce processes data (used in Exp 6).
- Single-node lab: replication = 1. fs.defaultFS = hdfs://localhost:9000

Hadoop on this Mac

- Java 17: brew install openjdk@17
- Hadoop 3.4.1 at /Users/ojas/hadoop-3.4.1 (not inside DS Lab folder)
- Each new terminal: export JAVA_HOME, HADOOP_HOME, PATH (see Hadoop_Download_Steps.pdf)
- Format once per fresh run: hdfs namenode -format
- Start: hdfs --daemon start namenode; hdfs --daemon start datanode
- Check: jps shows NameNode and DataNode; hdfs dfsadmin -report shows live DataNodes

Linux vs Hadoop commands

ls, mkdir, cat, cp, rm on your Mac disk are normal shell commands. hdfs dfs -ls, -mkdir, -cat, -cp, -rm work on HDFS. put uploads local to HDFS; get downloads HDFS to local. cp is source first, then destination.

Commands to remember

Command	Purpose
hdfs dfs -mkdir -p /path	Create folder
hdfs dfs -put file /path/	Upload
hdfs dfs -ls /path	List
hdfs dfs -cat /path/file	Show contents
hdfs dfs -cp src dest	Copy inside HDFS
hdfs dfs -get hdfs local	Download
hdfs dfs -rm -r /path	Delete folder
hdfs dfs -df -h /	Space summary
hdfs dfsadmin -report	Cluster health, live DataNodes

Exercises

Q: 1. New directory and multiple files.

A: mkdir -p /user/lab/exercise1; put file1.txt file2.txt; ls.

Q: 2. Copy between HDFS folders.

A: cp /user/lab/exercise1/file1.txt /user/lab/exercise2/file1_copied.txt

Q: 3. Delete a directory.

A: hdfs dfs -rm -r /user/lab/to_delete

Q: 4. Display file contents.

A: hdfs dfs -cat /user/lab/exercise1/file2.txt

Q: 5. Check space.

A: hdfs dfs -df -h / and hdfs dfsadmin -report

Viva questions

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Q: NameNode vs DataNode?

A: NameNode: metadata. DataNode: actual blocks.

Q: What is a block?

A: Fixed-size chunk of a file (default 128 MB).

Q: What does dfs mean?

A: Distributed file system; hdfs dfs runs file ops on HDFS.

Q: put vs get?

A: put: local to HDFS. get: HDFS to local.

Q: Why format NameNode?

A: Creates empty namespace; only once per new setup; wipes HDFS if repeated.

Q: What does dfsadmin -report do?

A: Shows capacity and whether DataNodes are alive.

Experiment 6: Word Count (MapReduce)

Aim

Count word frequencies in a text file using a Java MapReduce program on HDFS.

Original input.txt (3 lines)

Hadoop is a big data framework.
Hadoop uses MapReduce.
MapReduce processes big data efficiently.

- 14 words, 10 unique. hadoop, big, data, mapreduce count 2 each.

input_modified.txt

Same 3 lines plus: 'HDFS stores big data in blocks.' and 'Hadoop is open source and Hadoop is scalable.' -> 28 words, 18 unique.

What you did

- Upload input to HDFS. Compile WordCount.java to wordcount.jar.
- hadoop jar wordcount.jar WordCount <input> <output>
- Read hdfs dfs -cat <output>/part-r-00000

MapReduce flow

- Mapper: each line -> (word, 1)
- Combiner: local sum (same class as Reducer)
- Shuffle/Sort: group by word
- Reducer: sum counts -> (word, total)
- Output must be a new directory; delete old output with hdfs dfs -rm -r

Exercises

Q: 1. Modify input.

A: 28 words, 18 unique; hadoop 2 to 4.

Q: 2. Skip stopwords.

A: -D wordcount.skip.stopwords=true; map output 28 to 22.

Q: 3. Another file.

A: another.txt: 39 words, 29 unique.

Q: 4. Compare outputs.

A: Some words unchanged; some increased; new words (hdfs, blocks, ...).

Q: 5. Mapper and Reducer roles.

A: Mapper emits (word,1) per line; Reducer sums for final count.

Viva questions

Q: What is MapReduce?

A: Map builds key-value pairs; Reduce aggregates by key.

Q: Combiner?

A: Mini-reducer on map side; less network traffic.

Q: Shuffle and Sort?

A: Groups all values for each key before reduce.

Q: Why Text, IntWritable?

A: Hadoop serializable types for cluster I/O.

Q: part-r-00000?

A: Output file from reducer 0.

Q: _SUCCESS?

A: Marker that job completed.